

LLM Output Detectability and Task Performance Can be Jointly Optimized

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Abstract

Detecting machine-generated text is essential for transparency and accountability when deploying large language models (LLMs). Among detection approaches, watermarking is a statistically reliable method by design—it embeds detectable signals into LLM outputs by biasing their token distributions. However, it has been reported that watermarked LLMs often perform worse on downstream tasks. We propose **PUPPET**, a framework that fine-tunes an LLM via reinforcement learning to generate text that is both more detectable and better performing on downstream tasks. We use two reward functions: a detector that outputs a machine-class likelihood and an evaluator that measures a task-specific metric. Experiments on long-form QA, summarization, and essay writing show that LLMs trained with PUPPET achieve high detectability competitive with watermarking methods while outperforming them on downstream tasks. The analysis shows that this optimization can be performed efficiently with only a few thousand samples in 1–2 GPU hours. Moreover, these gains are consistent across out-of-domain tasks, different LLM families, and model sizes, and are even robust to paraphrasing attacks.

1 Introduction

LLMs now generate so fluent text that humans are almost unable to distinguish it from human-written text (Wang et al., 2025b; Clark et al., 2021; Jakesch et al., 2023). This raises urgent concerns about misuse, including academic fraud, spam, and automated disinformation campaigns (Chen & Shu, 2024; Goldstein et al., 2023). Therefore, detecting machine-generated text is essential for transparency and accountability.

From a LLM provider’s perspective, it is ideal for LLMs to be both highly detectable and strong on downstream tasks: high detectability supports accountability and misuse tracking, while strong task performance prevents users from exploring less detectable alternatives. *Watermarking* (Kirchenbauer et al., 2023) is the de facto standard approach toward this goal—by biasing token distributions at generation time, it enables statistically reliable detection via a hypothesis test with a provably controlled false-positive rate. However, since this approach optimizes only detectability, the resulting output distribution deviates from what is optimal for downstream tasks, leading to task performance degradation (Ajith et al., 2024; Fu et al., 2024; Wang et al., 2025a; Fernandez et al., 2023). Directly optimizing the LLM for both detectability and task performance simultaneously would address this limitation.

In this paper, we propose **PUPPET** , a framework for training an LLM via Direct Preference Optimization (DPO; Rafailov et al. (2023)) to produce texts that are both more detectable and better performing on downstream tasks. We use two reward functions: a detector that outputs a machine-generated class likelihood, and an evaluator that measures task performance. The detector encourages outputs that exhibit features indicative of machine-generated text, while the evaluator encourages the model to produce outputs that improve

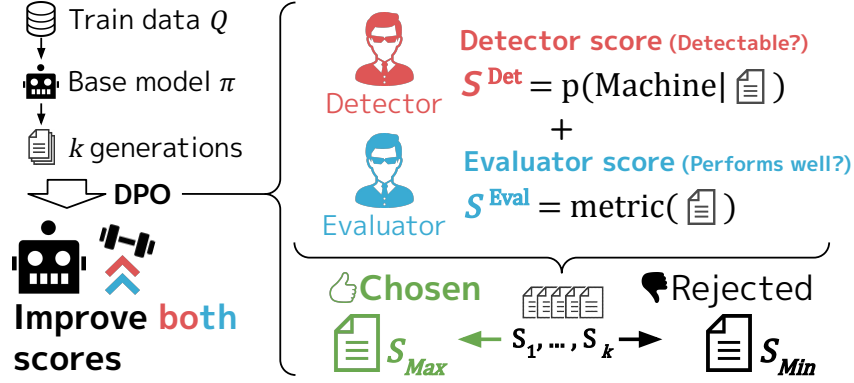


Figure 1: Overview of PUPPET. For each input query $q \in Q$, PUPPET samples k responses from the base model π , scores each with a detector score S^{Det} and an evaluator score S^{Eval} , assign “chosen” and “rejected” based on the combined score S , and fine-tunes π via DPO.

task performance. By constructing preference data from a combined score of these rewards and applying DPO, we aim to obtain a model that simultaneously improves both objectives.

Experiments on Llama-3-8B-Instruct (Grattafiori et al., 2024) across long-form QA, summarization, and essay generation show that PUPPET achieves comparable or superior detectability while consistently higher task performance, compared to five state-of-the-art watermarking methods: KGW (Kirchenbauer et al., 2023), SynthID (Dathathri et al., 2024), Unigram (Zhao et al., 2024), EXPGumbel (Aaronson, 2023), K-SemStamp (Hou et al., 2024b). For instance, on long-form QA, PUPPET improves both detectability and task performance by +3.8 and +2.6 points on average over those methods.

Importantly, these gains generalize across out-of-domain tasks and different base LLMs and detectors, and remain robust even to paraphrasing attacks (Krishna et al., 2023)—a major challenge for watermarking methods. In addition, PUPPET is practically efficient, requiring only a few thousand samples and 1–2 GPU hours. Finally, our exploratory analysis suggests that the learned detection-relevant features may also support model attribution, i.e., identifying whether a given LLM has been trained with PUPPET.

For reproducibility, our PUPPET implementation is publicly available on GitHub¹.

2 Proposed Method: PUPPET

Figure 1 illustrates an overview of PUPPET. We build a preference dataset and apply DPO to the model π , jointly optimizing both detectability and task performance. We adopt DPO for its training stability compared with policy-gradient alternatives such as PPO (Schulman et al., 2017). We describe the details of the construction of the preference data in this section.

First, we sample k responses t_j ($j \in \{1, \dots, k\}$) from the base model π for each input query q drawn from a query dataset Q (this dataset is task-specific):

$$\{t_1, \dots, t_k\} \sim \pi(\cdot | q).$$

To select “chosen” and “rejected” responses from the candidates, we design two scores to assess the detectability and task performance of a generated text t_j . The first is the **detector score**: the probability of the machine-generated class computed by a detector,

$$S_j^{\text{Det}} = p(\text{Machine} | t_j).$$

The second is the **evaluator score**: a task-specific evaluation score assigned by an evaluator,

$$S_j^{\text{Eval}} = \text{metric}(t_j),$$

¹<https://github.com/pakapaka333/PUPPET>

where $\text{metric}(\cdot)$ is a task-specific evaluation metric such as ROUGE-L (Lin, 2004) or LLM-as-a-Judge (Zheng et al., 2023).

The two scores are then combined into a final score for t_j ,

$$S_j = \alpha \cdot z(S_j^{\text{Det}}) + (1 - \alpha) \cdot z(S_j^{\text{Eval}}),$$

where $z(\cdot)$ denotes z-score normalization within the k samples to prevent either score from dominating due to differences in scale or variance, and $\alpha \in [0, 1]$ is a hyperparameter controlling the relative emphasis between detectability and task performance. Setting $\alpha > 0.5$ prioritizes detectability; and $\alpha < 0.5$ prioritizes task performance.

Finally, we build a dataset for DPO, $\{(q, t_{j^+}, t_{j^-}) \mid q \in Q\}$ by selecting a chosen response t_{j^+} and rejected response t_{j^-} for each query $q \in Q$,

$$j^+ = \operatorname{argmax}_{j \in \{1, \dots, k\}} S_j, \quad j^- = \operatorname{argmin}_{j \in \{1, \dots, k\}} S_j.$$

3 Experimental Setup

Datasets Q We use three datasets spanning diverse text-generation tasks: ELI5 (long-form QA; Fan et al. (2019); Guo et al. (2023)) and Multi-News (summarization; Fabbri et al. (2019)) from WaterBench (Tu et al., 2024), which is a standard benchmark for evaluating watermarking methods, and IELTS² (essay generation), which provides a long-form, open-ended setting. For each dataset, we randomly sample 5,000 instances for training and 200 for evaluation without overlap.

Base Model π We use Llama-3-8B-Instruct³ (Grattafiori et al., 2024) as the primary base model for all main experiments. To assess the robustness to the choice of base model, we additionally evaluate on the Qwen3 model family (QwenTeam, 2025) in Section 5.2.

Detector We use the OpenAI Detector (Solaiman et al., 2019) as the primary detector because it is widely used in prior work on machine-text detection (Nicks et al., 2024; Kushnareva et al., 2024). Additionally, we evaluate robustness to alternative classifiers in Section 5.2. We use AUROC as the evaluation metric for detection performance.

Evaluator To investigate the effect of metric characteristics, we use ROUGE-L (reported as a percentage in $[0, 100]$), a lexical overlap metric, for ELI5 and Multi-News. For IELTS, we use an LLM-as-a-Judge score on a $[0, 9]$ IELTS band scale to evaluate semantic quality.

Training / Evaluation We fine-tune LLMs using DPO with LoRA (Hu et al., 2022) for memory efficiency. We use a single fixed set of hyperparameters across all experiments for a controlled comparison, including $k = 5$ candidate responses (following Meng et al. (2024)) and $\alpha = 0.5$. For evaluation, models generate responses equal in number to the human responses per sample (ELI5: 3, Multi-News: 1, IELTS: 1) to ensure fair metric computation.

Baselines We compare PUPPET with five watermarking methods. Two are logit-based: **KGW** (Kirchenbauer et al., 2023), which biases green-list token logits conditioned on the preceding token, and **Unigram** (Zhao et al., 2024), which uses a fixed context-independent green list. Three are sampling-based: **EXPGumbel** (Aronson, 2023), which resamples tokens via a seeded Gumbel distribution; **SynthID** (Dathathri et al., 2024), which selects tokens through a tournament over multiple random scores; and **K-SemStamp** (Hou et al., 2024b), which steers sentence-level generation toward a watermarked region in semantic embedding space. Details of algorithms are provided in Section 6. We also report results for

²<https://github.com/chillestt/Automated-IELTS-essay-evaluation>

³Unless otherwise noted, “Llama-3” refers to “Llama-3-8B-Instruct” and “Qwen3” refers to “Qwen3-8B” throughout this paper.

Method	ELI5		Multi-News		IELTS	
	AUROC	ROUGE-L	AUROC	ROUGE-L	AUROC	Judge
WATERMARKING BASELINES						
KGW	99.6	23.5 ⁺	99.7	26.6	99.9	6.75
SynthID	93.5	23.5 ⁺	96.5	26.5	99.3	6.76
Unigram	93.1	23.2	94.5	26.7	98.8	6.73
EXPGumbel	98.7	22.6	94.4	26.5	98.2	4.81 [‡]
K-SemStamp	96.2	22.6	96.7	25.5 [‡]	96.5	6.73
OURS						
Vanilla	97.1	22.9	92.8	26.5	87.0	6.77
PUPPET 	100.0	25.8⁺	99.8	28.0⁺	99.3	7.19⁺

Table 1: In-domain detection (AUROC) and task performance (ROUGE-L, Judge) for Llama-3. Bold: best per column. ⁺ and [‡] denote significant improvement and degradation on task performance (paired t -test, $\alpha = 0.05$), respectively.

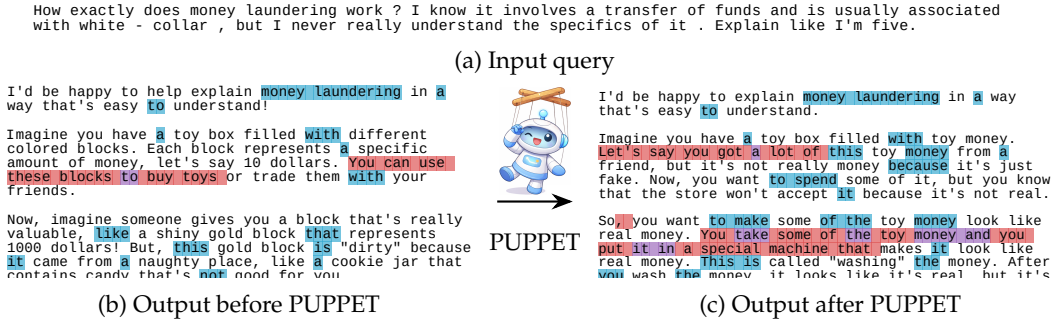


Figure 2: Spans that OpenAI Detector especially relies on (red) and spans contributing to ROUGE-L (LCS; blue), before and after PUPPET on a long-form QA example.

Vanilla baseline, the base model without PUPPET fine-tuning, evaluated with the OpenAI detector as a lower-bound reference.

For additional details on training and evaluation settings, see Appendix A.

4 Results

Table 1 presents the detection performance (AUROC) and task performance (ROUGE-L, Judge) of Llama-3 models trained with PUPPET across three benchmarks. We also visualize the token-level behavior of the responses using an ELI5 example in Figure 2. We draw two key observations.

PUPPET improves both detection and task performance simultaneously. Compared to Vanilla, PUPPET improves AUROC by up to +12.3 points (IELTS: 87.0 $\rightarrow$ 99.3), ROUGE-L by up to +2.9 points (ELI5: 22.9 $\rightarrow$ 25.8), and LLM-as-a-Judge score by up to +0.42 points (IELTS: 6.77 $\rightarrow$ 7.19). This confirms that PUPPET jointly optimizes both objectives, improving each relative to the Vanilla baseline.

PUPPET outperforms watermarking baselines in task performance while matching their detectability. Llama-3 models trained with PUPPET achieve task performance levels that watermarked models cannot, with a gap of up to 6% on IELTS (SynthID: 6.76 vs. PUPPET: 7.19), while matching or surpassing them on detection. For example, PUPPET outperforms KGW, the strongest watermarking baseline on detection, on 2 out of 3 benchmarks.

Train \ Eval	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
Vanilla	97.1	22.9	92.8	26.5	87.0	6.77
ELI5	100.0	25.8[†]	97.3	27.6 [‡]	94.9	6.64
MN	99.8	24.5 [‡]	99.8	28.0[†]	97.5	6.44
IELTS	99.6	24.2 [†]	97.5	27.0 [‡]	99.3	7.19[†]

Table 2: Detection and task performance for Llama-3. Rows: task used for PUPPET training; columns: evaluation benchmarks. Boxes highlight out-of-domain evaluations (off-diagonal). [†] and [‡] denote significant improvement and degeneration on task performance (paired t -test, $\alpha = 0.05$), respectively.

Model	Method	3-task avg.	
		Det.	Task
Qwen3-4B	Vanilla	88.7	44.2
	PUPPET	97.1	45.3
Qwen3-8B	Vanilla	81.2	44.4
	PUPPET	94.1	46.0
Qwen3-14B	Vanilla	76.0	45.2
	PUPPET	97.4	47.1

Table 3: Detection (Det.: AUROC) and task performance (Task: composite score), averaged over three benchmarks after aligning their scales, for Qwen3 variants of different sizes.

Figure 2 shows the learned behavior. We highlight two types of spans: red spans strongly contribute to detection, and blue spans contribute to task performance. Concretely, red spans are identified by SHapley Additive exPlanations (SHAP; (Lundberg & Lee, 2017))—a token-level attribution method that quantifies each span’s contribution to the OpenAI Detector’s output—and blue spans are identified via Longest Common Subsequence (LCS) with the reference, reflecting ROUGE-L overlap. See Appendix A.5 for full implementation details. After applying PUPPET, both span coverages increase: red by 1.8% and blue by 13%. This provides intuitive evidence that DPO-based PUPPET leads the base LLM to acquire features that benefit both detectability and task performance, as intended.

Note that the task performance degradation for watermarked models observed in prior work is not consistently reproduced in our experimental settings. We conjecture that this may be because recent LLMs can generate fluent text even under distributional constraints, though we do not conduct a dedicated analysis to substantiate this claim. Regardless, PUPPET’s contribution holds even in the absence of the reported degradation: PUPPET actively improves task performance to levels that watermarking methods cannot reach.

5 Analysis

We conduct five analyses to further characterize PUPPET: (1) *generalizability*—whether PUPPET transfers to out-of-domain tasks, (2) *robustness*—whether PUPPET holds up across the different choices of base LLM, detector, and paraphrasing attacks, (3) *efficiency*—whether PUPPET can be applied with modest data resources, (4) *reward ablation*—whether both reward components are necessary for jointly optimizing both objectives, and (5) *model attribution*—whether PUPPET enables distinguishing its outputs from those of other models.

5.1 Generalizability to Out-of-Domain Tasks

To assess whether PUPPET transfers beyond the tasks used during training, we evaluate the trained models on tasks not seen during training. Table 2 reports the out-of-domain (OOD) results (off-diagonal cells) alongside in-domain (ID) results (diagonal cells).

The results show that, even when evaluated on OOD tasks, PUPPET consistently improves detection performance relative to the Vanilla baseline without degrading task performance. For example, a model trained only on ELI5 improves AUROC on Multi-News from 92.8 to 97.3 and on IELTS from 87.0 to 94.9. This suggests that the detection-relevant features learned by PUPPET transfer across tasks, rather than overfitting to the ID distribution.

Detector	Method	3-task avg.	
		Det.	Task
OpenAI	Vanilla	92.3	41.5
	PUPPET	99.7	44.5
FakeSpotAI	Vanilla	99.9	41.5
	PUPPET	100.0	44.2
MAGE	Vanilla	95.5	41.5
	PUPPET	96.5	45.0

Table 4: Detection (Det.: AUCROC) and task performance (Task: composite score) averaged over three benchmarks after aligning their scales, across three detectors.

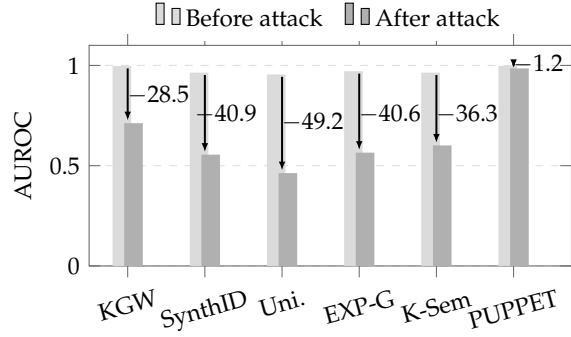


Figure 3: AUROC before and after Dipper paraphrasing (averaged over three benchmarks for Llama-3). Arrows and numbers indicate the drop in AUROC.

5.2 Robustness

We assess PUPPET’s robustness along three axes: (i) the choice of base LLM, (ii) the choice of detector, and (iii) adversarial paraphrasing attacks.

5.2.1 Base LLM Choice.

Table 3 reports results for three Qwen3 variants (QwenTeam, 2025) and examines robustness to LLM choice along two sub-axes: model family and model size.

Model family. Qwen3-8B is comparable in model size to Llama-3-8B-Instruct but belongs to a different model family, allowing us to isolate the effect of model family. PUPPET improves detection performance from 81.2 to 94.1 and task performance from 44.4 to 46.0 on average, demonstrating that its effectiveness generalizes beyond a single LLM family.

Model size. We further evaluate Qwen3-4B and Qwen3-14B to assess PUPPET’s sensitivity to model size.⁴ PUPPET consistently improves both objectives across all model sizes. The detection gain is particularly large for Qwen3-14B (76.0 → 97.4), where the stronger base model is intrinsically harder to detect, yet PUPPET still closes most of the gap. On the task side, PUPPET further improves each Vanilla baseline, regardless of size, demonstrating that PUPPET complements rather than undermines the capabilities of larger models.

These results confirm that PUPPET’s effectiveness is not contingent on any particular model family or model size. Full per-task results for the Qwen3 family are provided in Appendix B.

5.2.2 Detector Choice.

Table 4 shows results with FakeSpotAI⁵ and MAGE(Li et al., 2024), two detectors not used in the main results, alongside the OpenAI Detector for reference. PUPPET consistently improves both objectives across all detectors. Notably, FakeSpotAI already achieves high detection performance on the Vanilla baseline, yet PUPPET further improves it while also lifting task performance. This demonstrates that PUPPET’s gains are not confined to specific detectors. Full per-task results are provided in Appendix C.

5.2.3 Paraphrasing Attacks.

We evaluate robustness to paraphrasing attacks with Dipper (Krishna et al., 2023). Dipper is a widely used paraphrasing attack model (Zhao et al., 2024; Rastogi & Pruthi, 2024; Liu

⁴Llama-3-70B-Instruct is the only choice for a different size variant of the Llama-3 series. However, it is too large to train under our computational resources, so we chose Qwen3 variants instead.

⁵<https://huggingface.co/fakespot-ai/roberta-base-ai-text-detection-v1>

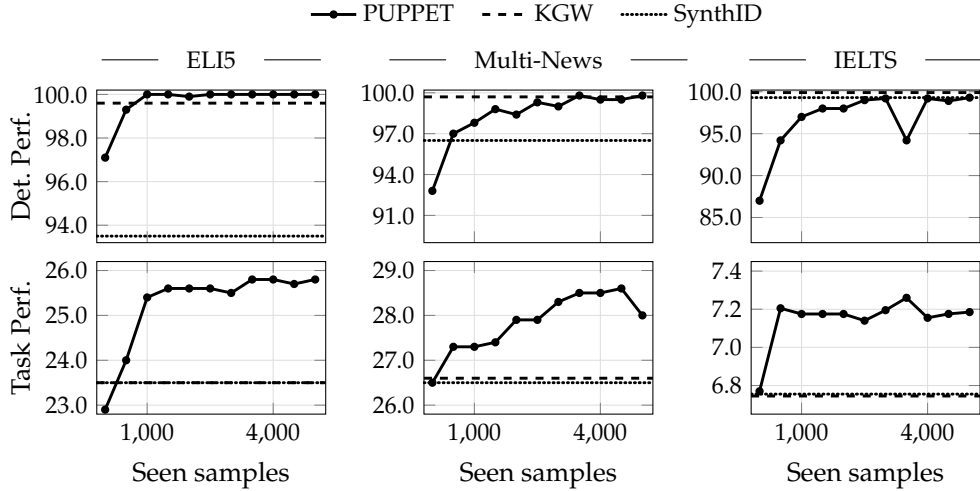


Figure 4: Detection performance (Det. Perf., AUROC; top row) and task performance (Task Perf., ROUGE-L for ELI5 and Multi-News, judge score for IELTS; bottom row) as a function of training samples seen for Llama-3 across all three benchmarks (500 samples per dot). Dashed and dotted lines indicate KGW and SynthID baselines, respectively.

et al., 2024) that rewrites text at the surface level while preserving meaning. Figure 3 reports AUROC averaged over the three benchmarks for Llama-3 models. Full per-task results are provided in Appendix D.1. While all watermarking baselines suffer severe degradation—up to a 49.2-point drop in AUROC—PUPPET’s AUROC decreases by only 1.2 points on average. We attribute this near-zero degradation to the learned detection-relevant features, which are expected to go beyond surface-level statistical patterns in the token distribution. If models trained with PUPPET successfully internalize those features—such as detection-relevant sentence structures and stylistic patterns—the features would be difficult for surface-level paraphrasing to remove. We also evaluate using FakeSpotAI and MAGE and find that the same trend holds; these results are reported in Appendix D.2.

5.3 Training Efficiency

Even a method that achieves strong performance may have limited practical value if it demands prohibitive data or computational resources. We therefore examine PUPPET’s practicality along both dimensions. Figure 4 plots AUROC and task performance as a function of the number of training samples seen for Llama-3 across all three benchmarks.

Across all benchmarks, both detectability and task performance rise sharply within the first few thousand samples and plateau near the level achieved with the full training set. For example, on ELI5, both performances converge within 1k–2k samples, with AUROC already reaching 100.0 at 1k samples. This handful of DPO pairs represents a remarkably small budget compared to large-scale DPO-based alignment pipelines such as OLMo3 (Team Olmo, 2025), which train on 150K–200K pairs. Regarding compute, each training run completes in approximately 1–2 hours on a single NVIDIA A6000 (48GB) GPU. These results confirm that PUPPET is both data-efficient and computationally lightweight. The same trend holds for Qwen3 (Appendix E).

5.4 Reward Composition Ablation

PUPPET uses a composite reward comprising a detector score (S^{Det}) and an evaluator score (S^{Eval}). To assess the necessity of this combined design, we compare PUPPET ($\alpha = 0.5$) against single-reward variants: S^{Det} only ($\alpha = 1$) and S^{Eval} only ($\alpha = 0$), which serve as

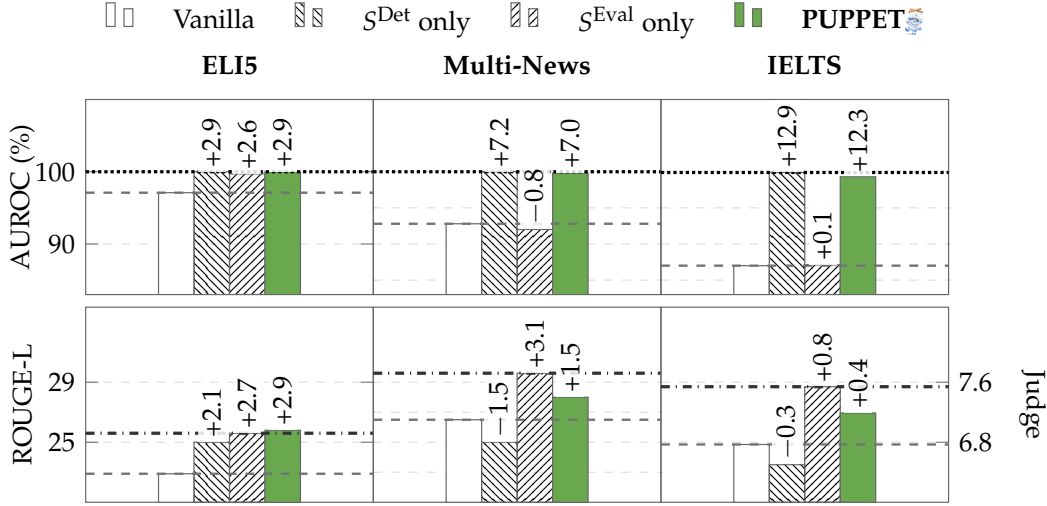


Figure 5: Reward ablation (Llama-3). Bar labels: point-wise changes from Vanilla. In the AUROC panels, dotted lines mark the S^{Det} -only ceiling; in the task panels, dash-dotted lines mark the S^{Eval} -only ceiling; dashed lines mark the Vanilla baseline in all panels. ROUGE-L (left) and Judge (right) task axes are proportionally scaled.

Target	Contrast	ELI5	Multi-News	IELTS
Vanilla Llama-3	Vanilla Qwen3	58.99 (0.23)	72.26 (0.82)	77.71 (0.96)
PUPPET Llama-3	Vanilla Qwen3	94.47 (1.77)	96.94 (3.18)	99.14 (4.62)
PUPPET Llama-3	Vanilla Llama-3	92.18 (1.45)	87.32 (1.67)	94.04 (2.07)
PUPPET Llama-3	PUPPET Qwen3	54.54 (0.22)	81.64 (1.21)	87.22 (1.13)

Table 5: Model-attribution AUROC (OpenAI Detector), treating *Target* as the positive class and *Contrast* as the negative. Parentheses: Cohen’s d ; models are trained per task.

single-objective upper bounds for detectability and task performance, respectively. Figure 5 shows Llama-3 results; bar labels report signed point-wise changes from the Vanilla baseline.

Optimizing detectability alone (S^{Det} only) achieves large AUROC gains over Vanilla (up to +12.9 points) but degrades task performance (up to -1.5 ROUGE-L and -0.3 Judge). Conversely, optimizing task performance alone (S^{Eval} only) yields meaningful improvements in it (up to +3.1 ROUGE-L and +0.8 Judge) but provides no reliable detection gain. PUPPET jointly optimizes both objectives and recovers 95.3–100% of the S^{Det} -only detection gain while simultaneously improving task performance over Vanilla. These results confirm that both reward components are necessary: neither alone achieves strong performance across both objectives. Moreover, α provides a practical knob: higher values favor detectability, while lower values favor task performance. The same trend holds for Qwen3 (Appendix F).

5.5 PUPPET Enables Model Attribution

As an exploratory analysis, we investigate the use of PUPPET for model attribution. PUPPET is expected to elevate the machine-generated class likelihood of the base model, enabling us to distinguish the trained model from other models by setting a sufficiently high threshold on the likelihood. We evaluate this by treating *Target* model outputs as the positive (machine) class and *Contrast* model outputs as the negative (human) class, computing AUROC from the OpenAI Detector’s machine-generated class likelihood. Table 5 shows Llama-3 results.

Attribution accuracy (AUROC) is governed by the gap in machine-generated class likelihood between the two models; we quantify this with Cohen’s d , reported alongside AUROC in

Table 5. At the Vanilla baseline, the gap is small on ELI5 ($d = 0.23$), yielding near-chance attribution (AUROC 58.99), and grows on Multi-News and IELTS, supporting only moderate performance (72.26–77.71). After PUPPET fine-tuning, the gap widens substantially (up to $d = 4.62$ on IELTS) and attribution rises markedly to 94.47–99.14, confirming that PUPPET’s controlled upward shift creates a reliable threshold.

As a control, we consider the case where both models are trained with PUPPET. On ELI5, where both models achieve near-perfect AUROC (100.0 and 99.9; Tables 1 and 8), the two PUPPET models converge to the same likelihood region ($d = 0.22$), and attribution collapses to near chance (54.54); on Multi-News and IELTS, where their distributions remain more separated ($d = 1.21, 1.13$), attribution stays above chance (81.64 and 87.22).

We treat these findings as preliminary evidence of PUPPET’s potential for model attribution. The similar trend holds for Qwen3 (Appendix G).

6 Related Work

Research on machine text detection can be broadly categorized into two directions: (1) detectors: utilizing discriminative features of machine-generated text, and (2) watermarking: embedding detectable features into text during generation.

Detectors A standard approach involves training supervised classifiers on labeled datasets of human-written and machine-generated text. RoBERTa (Liu et al., 2019) is widely used as the base model (Solaiman et al., 2019; Wang et al., 2024b; Tulchinskii et al., 2023; Li et al., 2024), and the RoBERTa-based OpenAI Detector (Solaiman et al., 2019) is widely used as a benchmark in prior work (Nicks et al., 2024; Kushnareva et al., 2024).

A complementary, training-free paradigm exploits statistical properties of the generation process. DetectGPT (Mitchell et al., 2023) exploits the observation that machine-generated text lies near local maxima of a model’s log-probability surface, so perturbations lower its log-probability, whereas human-written text does not exhibit this behavior. Fast-DetectGPT (Bao et al., 2024) achieves comparable accuracy at substantially lower cost by replacing perturbation-based estimation with conditional probability curvature sampling. Binoculars (Hans et al., 2024) achieves strong zero-shot detection by computing the ratio of cross-perplexity and perplexity between two related models.

Watermarking Watermarking methods embed detectable signals into LLM-generated text and can be grouped into logit-based and sampling-based approaches.

Logit-based methods bias the token output distribution at generation time. KGW (Kirchenbauer et al., 2023) partitions the vocabulary into green and red lists conditioned on the preceding token, and increases the logits of green-list tokens by a fixed δ . Unigram (Zhao et al., 2024) simplifies this by using a fixed, context-independent green list, thereby making it more robust to token-substitution attacks. Both methods introduce systematic distortions into the output distribution, which has been shown to degrade task performance (Ajith et al., 2024; Fu et al., 2024; Wang et al., 2025a; Fernandez et al., 2023).

Sampling-based methods, in contrast, intervene at the token selection stage without directly modifying logits. EXPGumbel (Aaronson, 2023) resamples tokens via a seeded Gumbel distribution, and SynthID (Dathathri et al., 2024) selects tokens through a tournament procedure over multiple random scores. K-SemStamp (Hou et al., 2024b) is a clustering-based extension of SemStamp (Hou et al., 2024a), which steers sentence-level generation toward watermarked regions in semantic embedding space. These methods are generally less disruptive to the output distribution than logit-based approaches, but task performance degradation can still be observed (Wang et al., 2025a).

7 Conclusion

We presented PUPPET, the DPO-based framework that trains an LLM to produce text that is both highly detectable and performant on downstream tasks. By jointly optimizing detectability and task performance, PUPPET matches watermarking methods in detection while surpassing them on downstream tasks—a combination that prior methods, which optimize only one objective at a time, cannot achieve. These gains generalize to out-of-domain tasks and prove robust to the choice of base LLM, detector, and paraphrasing attacks, all within moderate data and computational resources. Our exploratory analysis further suggests that PUPPET opens the door to model attribution.

Two directions remain for future work: deeper mechanistic explanations of the acquired detect-relevant features, and systematic hyperparameter tuning—including the relative weighting of the two rewards—which may yield further gains. We hope this work encourages research into jointly optimizing LLMs across multiple accountability objectives, thereby contributing to more transparent deployment of language models.

Ethics Statements

Detection approaches, including PUPPET and watermarking, are still in the research stage. Therefore, it is essential to use these techniques with utmost caution under high-stakes settings where false positives (incorrectly identifying human-written text as machine-generated text) may cause significant harm.

Reproducibility Statements

We release the PUPPET implementation on GitHub. The repository contains the experimental code and configuration files used in this paper, enabling researchers to reproduce our results. It also provides a refactored implementation to facilitate applying PUPPET to researchers’ own models, data, and tasks.

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Hyperparameter	Value
Learning rate	1×10^{-4}
Epochs	1
Batch size (per device)	ELI5: 20; Multi-News: 5; IELTS: 10
Data seed	42
Max sequence length	ELI5: 800; Multi-News / IELTS: 8,192
Max completion length	ELI5: 300; Multi-News / IELTS: 512

Table 6: Training hyperparameters. Values not listed use TRL / PEFT defaults.

Task	Split	Source	Samples
ELI5	Train	Hello-SimpleAI/HC3, reddit_eli5	5,000
	Eval	THU-KEG/WaterBench (2-1_longform_qa)	200
Multi-News	Train	alexabbri/multi_news	5,000
	Eval	THU-KEG/WaterBench (4-1_multi_news)	200
IELTS	Train	chillies/IELTS-writing-task-2-evaluation	5,000
	Eval	chillies/IELTS-writing-task-2-evaluation	200

Table 7: Data sources for training and evaluation splits.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. URL <https://openreview.net/forum?id=ucCHPGDlao>.

A Implementation Details

A.1 Generation Settings

All text generation—both for constructing DPO preference data and for evaluation—uses vLLM (Kwon et al., 2023) (v0.11.0). To evaluate instruct-tuned models correctly, we apply chat templates. All parameters related to generation use defaults, except for `n` (number of responses per prompt; see Section 3) and `enable_thinking` (False for all Qwen3 experiments, to exclude any effects of thinking).

A.2 Training Setup

We fine-tune with the DPOTrainer in TRL⁶, using LoRA for memory efficiency. Table 6 lists the key hyperparameters; values not shown use TRL defaults. We adopt a single fixed configuration without systematic search.

A.3 Evaluation Setup

A.3.1 ROUGE-L

ROUGE-L for ELI5 and Multi-News follows the WaterBench (Tu et al., 2024) implementation, where we compute the ROUGE-L score for each response and adopt the best score per sample.

⁶<https://github.com/huggingface/trl>

A.3.2 LLM-as-a-Judge Evaluation

For IELTS, task performance is measured using an LLM-as-a-Judge score on a $[0, 9]$ IELTS band scale. We use GPT-OSS-20B (OpenAI, 2025) as the judge model with temperature 0.0 and `max_tokens` = 4096. The scoring prompt is adapted from Wang et al. (2024a).

Probability-weighted scoring. Rather than parsing the generated integer directly, we compute the final score as a probability-weighted expected value over the ten score labels $\{0, 1, \dots, 9\}$, following Liu et al. (2023):

$$\hat{s} = \sum_{v=0}^9 v \cdot P(v), \quad P(v) = \frac{\exp(\ell_v)}{\sum_{v'=0}^9 \exp(\ell_{v'})}, \quad (1)$$

where ℓ_v is the log-probability of score token v as assigned by the judge model. This yields a continuous score in $[0, 9]$ and, crucially, produces sufficient variance across the $k = 5$ preference candidates—a necessary condition for effective DPO training.

Handling the reasoning model. Because GPT-OSS-20B is a reasoning model, it generates a thinking trajectory before producing the final score token. To compute token-level log-probabilities conditioned on that reasoning, we proceed in two steps: (1) we first generate the full thinking trajectory by sampling with a stop sequence that marks the end of the thinking phase; (2) we then prepend the thinking trajectory and the prefix “Score: ” to each candidate score token and extract the corresponding prompt log-probability via vLLM. This ensures that ℓ_v reflects the model’s judgment after reasoning rather than being estimated unconditionally. Step (1) may fail to complete the thinking trajectory within the token limit (`max_new_tokens`). We set the maximum number of retries to 20 and exclude samples whose thinking trajectory fails to terminate.⁷

A.4 Watermark Baseline Setup

We adopt MarkLLM (Pan et al., 2024) to generate all watermarked texts (KGW, Unigram, EXPGumbel, SynthID, K-SemStamp), with the only modification being the application of chat templates. We use the default configurations bundled in MarkLLM.

A.5 Span Visualization

The span visualization in Figure 2 is produced using the Python shap library⁸, operating at the granularity of the model’s tokenizer.

Detector-salient spans (red). We compute SHAP values for the OpenAI Detector with respect to class 0 (Machine). Only the positive component of each token’s SHAP value is retained (`mode = "pos"`), capturing tokens that actively push the prediction toward the Machine class. To normalize scores comparably across conditions, we compute a shared upper bound $c_{\max}$ as the 99th percentile of the pooled positive SHAP scores from both the Vanilla and PUPPET outputs, ensuring that the color scale is directly comparable between the two conditions. Tokens whose normalized score meets or exceeds the threshold $\tau = 0.4$ are grouped into contiguous spans and highlighted in red.

Task-quality spans (blue). For ROUGE-L, task-relevant spans are identified via the longest common subsequence (LCS) between the generated text and the reference answer, computed at the word level (whitespace tokenization). Matched word spans are mapped back to character offsets and highlighted in blue. Extending this visualization to LLM-as-a-Judge

⁷LLM-as-a-Judge failures were negligible: 3 out of 25,000 judgments (5,000 samples $\times$ 5 responses) for Llama-3 and 1 out of 25,000 for Qwen3. The corresponding samples were excluded from preference pair construction.

⁸<https://github.com/maciejskorski/shap>

Method	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
WATERMARKING BASELINES						
KGW	99.4	24.5	99.3	24.1	99.9	7.78[†]
SynthID	94.5	24.6	98.0	24.2	98.9	7.67
Uni.	91.1	24.1 [‡]	97.0	24.3	99.1	7.68
EXP-G	98.8	23.7 [‡]	96.3	23.7	97.8	4.89 [‡]
K-Sem	98.6	24.7	98.5	23.4 [‡]	98.5	7.67
OURS						
Vanilla	93.7	24.5	83.3	24.0	66.6	7.63
PUPPET	99.9	26.6[†]	96.7	26.4[†]	85.6	7.65

Train	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
Vanilla	93.7	24.5	83.3	24.0	66.6	7.62
ELI5	99.9	26.6[†]	90.1	24.4	82.1	7.65
MN	98.6	25.4 [†]	96.7	26.4[†]	87.4	7.52
IELTS	98.5	25.3 [†]	89.0	24.6 [†]	85.6	7.65

(a) In-domain results for Qwen3-8B, including watermarking baselines.

(b) Out-of-domain results for Qwen3-8B. Each row shows the task PUPPET was trained on; columns show evaluation benchmarks. Boxes highlight off-diagonal (OOD) entries.

Table 8: Detection performance and task performance using Qwen3-8B, on the (a) in-domain evaluation set (left) and (b) out-of-domain evaluation set (right). Bold values indicate the best result in each column. Boxes in the right subtable highlight out-of-domain evaluations (off-diagonal). [†] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

spans proved computationally prohibitive (estimated >1 week), and we therefore report only ROUGE-L-based spans.

A.6 Paraphrasing Attack Setup

To evaluate robustness to adversarial paraphrasing, we apply Dipper (Krishna et al., 2023) to all model outputs before running detection. All experiments use identical Dipper settings across benchmarks and base models. We set `lex_diversity` and `order_diversity` to 60, following the strongest parameter values reported by Krishna et al. (2023). Other hyperparameters use defaults.

B Full Results for Other Base LLMs

This appendix provides full per-task results for all Qwen3 models evaluated in Section 5.2, organized by the robustness axis under investigation: Section B.1 addresses robustness to model family (Qwen3-8B vs. Llama-3-8B-Instruct), and Section B.2 addresses robustness to model size (Qwen3-4B, 8B, and 14B).

B.1 Robustness to Model Family: Qwen3-8B

Table 8 (left subtable) shows in-domain results for Qwen3-8B. On ELI5 and Multi-News, PUPPET markedly improves both AUROC and task performance compared to Vanilla, consistent with the Llama-3 results in Table 1. On IELTS, however, the AUROC gain is more modest ($66.6 \rightarrow 85.6$) compared to Llama-3 ($87.0 \rightarrow 99.3$). The lower Vanilla AUROC on IELTS (66.6 vs. 87.0 for Llama-3) suggests that Qwen3-8B’s essay-style outputs are further from the OpenAI Detector’s training distribution, making the detection signal harder to amplify via DPO. Despite this, PUPPET still achieves a meaningful absolute gain, confirming that the framework is not inherently limited to a single LLM family.

Table 8 (right subtable) shows out-of-domain results. As with Llama-3 (Table 2), PUPPET consistently improves detection over Vanilla even on tasks not seen during training, confirming that the detection-relevant features learned by PUPPET transfer across tasks regardless of the base model family.

Method	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
WATERMARKING BASELINES						
KGW	96.2	24.3	96.4	25.2	99.5	7.57
SynthID	86.8	24.5	93.5	25.1	97.3	7.60
Uni.	73.5	24.3	85.3	25.4 [†]	97.0	7.63[†]
EXP-G	88.4	23.8 [‡]	87.9	25.4	95.9	4.86 [‡]
K-Sem	96.1	24.3	95.4	24.2 [‡]	98.2	7.57
OURS						
Vanilla	98.2	24.5 [†]	88.0	24.9 [†]	79.9	7.50
PUPPET	100.0	25.6	95.4	26.7	96.0	7.53

(a) In-domain results for Qwen3-4B.

Train	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
Vanilla	98.2	24.5	88.0	24.9	79.9	7.50
ELI5	100.0	25.6[†]	94.4	25.6 [†]	89.7	7.39
MN	99.0	25.2 [†]	95.4	26.7[†]	89.2	7.45
IELTS	99.8	24.8 [†]	95.6	25.6 [†]	96.0	7.53

(b) Out-of-domain results for Qwen3-4B. Each row shows the task PUPPET was trained on; columns show evaluation benchmarks. Boxes highlight off-diagonal (OOD) entries.

Table 9: Detection performance and task performance using Qwen3-4B, on the (a) in-domain evaluation set (left) and (b) out-of-domain evaluation set (right). Bold values indicate the best result in each column. Boxes in the right subtable highlight out-of-domain evaluations (off-diagonal). [†] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

Method	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
WATERMARKING BASELINES						
KGW	99.2	24.4	99.7	24.3	99.8	7.88
SynthID	94.5	24.2	98.1	24.4	99.3	7.80
Uni.	90.7	24.0 [‡]	96.8	24.3	98.6	7.78
EXP-G	98.5	23.3 [‡]	96.9	24.3	98.1	4.98 [‡]
K-Sem	98.0	24.2	99.0	23.2 [‡]	99.0	7.84
OURS						
Vanilla	86.4	24.4	76.4	24.0	65.1	7.86
PUPPET	99.6	26.0[†]	99.1	28.3[†]	93.6	7.81

(a) In-domain results for Qwen3-14B.

Train	ELI5		MN		IELTS	
	AUC	R-L	AUC	R-L	AUC	Jdg.
Vanilla	86.4	24.4	76.4	24.0	65.1	7.86
ELI5	99.6	26.0[†]	82.3	24.7 [†]	80.6	7.73 [‡]
MN	98.5	25.4 [†]	99.1	28.3[†]	93.4	7.44 [‡]
IELTS	97.2	25.0 [†]	86.6	24.8 [†]	93.6	7.81

(b) Out-of-domain results for Qwen3-14B. Each row shows the task PUPPET was trained on; columns show evaluation benchmarks. Boxes highlight off-diagonal (OOD) entries.

Table 10: Detection performance and task performance using Qwen3-14B, on the (a) in-domain evaluation set (left) and (b) out-of-domain evaluation set (right). Bold values indicate the best result in each column. Boxes in the right subtable highlight out-of-domain evaluations (off-diagonal). [†] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

B.2 Robustness to Model Size: Qwen3-4B, 8B, and 14B

Tables 9 and 10 show results for Qwen3-4B and Qwen3-14B, respectively, complementing the Qwen3-8B results above.

Detection performance. A notable pattern across model sizes is that Vanilla detection AUROC decreases monotonically with parameter count (3-task averages: 88.7 for 4B, 81.2 for 8B, 76.0 for 14B), consistent with the intuition that larger models produce more fluent, human-like text that is intrinsically harder for the detector to identify. PUPPET not only reverses this trend but achieves consistently high AUROC across all sizes (97.1, 94.1, and 97.4 for 4B, 8B, and 14B, respectively). The absolute detection gain increases with model size (+8.4, +12.9, and +21.4 AUROC points), indicating that PUPPET compensates most strongly where Vanilla detection is weakest. As observed for Llama-3 in Table 1, PUPPET’s detection AUROC is competitive with watermarking baselines on both Qwen3-4B and Qwen3-14B, while consistently surpassing them on task performance (see Tables 9 and 10).

Task performance. Task performance improves consistently across the 4B and 8B models. For Qwen3-14B, ELI5 and Multi-News ROUGE-L both improve substantially ($24.4 \rightarrow 26.0$ and $24.1 \rightarrow 28.3$, respectively), while IELTS Judge decreases marginally ($7.86 \rightarrow 7.81$). This slight degradation in IELTS for the 14B model is the only case across all our experiments in which PUPPET does not simultaneously improve task performance. However, the drop is not statistically significant and is small relative to the detection gain.

Out-of-domain generalization. The OOD results (right subtables of Tables 9 and 10) show that cross-task generalization holds for both Qwen3-4B and Qwen3-14B: PUPPET trained on any single task consistently improves detection on the other two tasks over Vanilla.

C Full Results Across Detectors

This section presents full per-task detection and task performance results for all evaluated detectors, complementing the summary averaged over three tasks in Table 4. In all experiments reported here, each detector serves as both the reward signal during PUPPET training and the evaluation metric—no cross-detector transfer is assessed.

Table 11 reports the full results for FakeSpotAI and MAGE, covering both in-domain and out-of-domain generalization.

In-Domain. When trained and evaluated on the same task (diagonal entries in Table 11), PUPPET consistently improves or maintains detection performance relative to Vanilla across all detector-model combinations. Gains are most pronounced for MAGE on IELTS, where the Vanilla baseline is weakest: MAGE AUROC improves from 89.4 to 94.2 (Llama-3) and from 91.3 to 95.3 (Qwen3) after PUPPET training on IELTS. FakeSpotAI already achieves near-perfect detection under the Vanilla baseline, so in-domain PUPPET training maintains perfect detection (100.0) while consistently improving task performance. One exception is MAGE on Multi-News with Llama-3, which yields a small detection drop ($97.2 \rightarrow 95.4$, -1.8 points); this may reflect a mild tension between improving text quality and maintaining MAGE-detectable features.

Out-of-Domain. Cross-task generalization is strong for both FakeSpotAI and MAGE: models trained on any single task maintain high AUROC when evaluated on the other two, mirroring the pattern seen with the OpenAI Detector in Table 2. For FakeSpotAI, all off-diagonal AUROC values remain at or near 100.0 for both models, indicating that detection generalizes fully across tasks. MAGE also generalizes well, with most off-diagonal entries matching or exceeding the Vanilla baseline; the few small drops (at most -1.8 points) are consistent with the in-domain exception noted above and do not undermine the overall conclusion of robust cross-task transfer.

D Full Paraphrasing Attack Results

This appendix presents full paraphrasing attack results complementing the summary in Section 5.2. Section D.1 covers results when PUPPET is trained with the OpenAI Detector; Section D.2 examines the effect when alternative detectors are used for training.

D.1 Paraphrasing Attacks Against the OpenAI Detector

Table 12 reports AUROC and task performance before and after applying Dipper (Krishna et al., 2023) as a paraphrasing attack, across both models and all benchmarks. Each cell shows *Before / After* the attack.

For all watermarking baselines, AUROC drops sharply after paraphrasing, as surface-level token substitutions destroy the statistical patterns introduced at decoding time. In contrast, PUPPET’s AUROC is largely unaffected, and even after the attack, PUPPET consistently outperforms the Vanilla baseline in both detection and task quality—with in-domain results

Detector Train		ELI5		Multi-News		IELTS	
		AUROC	ROUGE-L	AUROC	ROUGE-L	AUROC	Judge
Llama-3							
Fake-SpotAI	Vanilla	100.0	22.9	100.0	26.5	99.7	6.77
	ELI5	100.0	25.2 [†]	100.0	27.1 [†]	99.8	6.74
	MN	100.0	23.9 [‡]	100.0	26.1	99.9	7.32
	IELTS	100.0	23.7 [†]	100.0	26.3	99.9	7.32 [†]
MAGE	Vanilla	100.0	22.9	97.2	26.5	89.4	6.77
	ELI5	100.0	24.5 [†]	97.3	27.2 [†]	90.0	6.66
	MN	100.0	23.5 [‡]	95.4	29.2 [†]	90.1	6.68
	IELTS	100.0	22.1 [‡]	96.7	26.0	94.2	7.31 [†]
Qwen3							
Fake-SpotAI	Vanilla	100.0	24.5	100.0	24.0	98.9	7.63
	ELI5	100.0	25.7 [†]	100.0	24.4 [†]	99.4	7.67
	MN	100.0	25.0 [‡]	100.0	24.1	99.4	7.64
	IELTS	100.0	24.9 [†]	100.0	24.0	99.6	7.63
MAGE	Vanilla	100.0	24.5	97.5	24.0	91.3	7.63
	ELI5	100.0	25.3 [†]	97.2	24.4	91.2	7.65
	MN	100.0	24.4	97.4	25.9 [†]	91.2	7.63
	IELTS	100.0	25.5 [†]	98.1	24.6 [†]	95.3	7.85 [†]

Table 11: Detection and task performance across alternative detectors and training datasets, for both Llama-3 and Qwen3. Detection is measured by AUROC, and task performance by ROUGE-L for ELI5 and Multi-News, or LLM-as-a-Judge score (Judge) for IELTS. Vanilla denotes the base model without PUPPET fine-tuning. Each PUPPET row indicates the single dataset on which PUPPET was trained. Boxes highlight out-of-domain evaluations (train task $\neq$ eval task). [†] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

again exceeding out-of-domain ones. This pattern suggests that PUPPET captures detection-relevant features that are not easily erased by surface-level rewriting, and is consistent across both Llama-3 and Qwen3.

Task performance declines after paraphrasing across all methods. This degradation reflects a property of Dipper itself: as a T5-based model, Dipper produces text of substantially lower quality than Llama-3 or Qwen3, causing semantic degradation regardless of the original generation method.

D.2 Paraphrasing Attacks Against Alternative Detectors

Table 13 reports in-domain detection and task performance before and after the Dipper paraphrasing attack for PUPPET trained with FakeSpotAI and MAGE. Only in-domain settings are evaluated; each PUPPET row includes results only for the task on which it was trained.

Detector-dependent AUROC change. Unlike the consistent pattern observed with the OpenAI Detector, AUROC changes after paraphrasing vary substantially across detectors. For FakeSpotAI, AUROC remains near-perfect both before and after paraphrasing for all PUPPET settings, as the Vanilla baseline already saturates at 100.0. For MAGE, AUROC after paraphrasing is comparable to or lower than the value before the attack in most settings, with notable drops on Multi-News (e.g., 95.4 $\rightarrow$ 84.9 for Llama-3). This variability likely reflects whether the text produced by Dipper retains features that each detector relies on: when a detector’s signal depends on surface-level patterns easily destroyed by rewriting, detection collapses; when those features are more abstract or incidentally preserved, detection holds. Crucially, for both FakeSpotAI and the OpenAI Detector, PUPPET consistently outperforms Vanilla after paraphrasing in detection and downstream tasks, and in-domain results exceed

Method	ELI5		Multi-News		IELTS	
	AUROC	ROUGE-L	AUROC	ROUGE-L	AUROC	Judge
Llama-3						
WATERMARKING BASELINES						
KGW	99.6 / 70.1	23.5 / 23.9 [†]	99.7 / 71.6	26.6 / 23.2 [‡]	99.9 / 72.0	6.75 / 5.45 [‡]
SynthID	93.5 / 55.7	23.5 / 23.6	96.5 / 54.3	26.5 / 23.0 [‡]	99.3 / 56.5	6.76 / 5.41 [‡]
Unigram	93.1 / 32.3	23.2 / 23.3	94.5 / 61.3	26.7 / 23.1 [‡]	98.8 / 45.3	6.73 / 5.26 [‡]
EXPGumbel	98.7 / 65.8	22.6 / 22.4	94.4 / 54.2	26.5 / 21.9 [‡]	98.2 / 49.4	4.81 / 4.84
K-SemStamp	96.2 / 60.6	22.6 / 23.6 [†]	96.7 / 65.1	25.5 / 22.9 [‡]	96.5 / 54.7	6.73 / 5.54 [‡]
OURS						
Vanilla	97.1 / 98.8	22.9 / 23.3 [†]	92.8 / 98.8	26.5 / 23.3 [‡]	87.0 / 92.9	6.77 / 5.46 [‡]
ELI5	100.0 / 100.0	25.8 / 24.9 [†]	97.3 / 98.9	27.6 / 23.9 [‡]	94.9 / 95.0	6.64 / 5.62 [‡]
MN	99.8 / 99.4	24.5 / 24.2 [‡]	99.8 / 99.4	28.0 / 24.5 [‡]	97.5 / 97.2	6.44 / 5.56 [‡]
IELTS	99.6 / 99.8	24.2 / 23.7 [†]	97.5 / 98.8	27.0 / 23.6 [‡]	99.3 / 96.2	7.19 / 5.97 [‡]
Qwen3						
WATERMARKING BASELINES						
KGW	99.4 / 59.2	24.5 / 24.4	99.3 / 60.9	24.1 / 22.9 [‡]	99.9 / 66.1	7.78 / 5.94 [‡]
SynthID	94.5 / 55.0	24.6 / 24.4	98.0 / 61.2	24.2 / 22.9 [‡]	98.9 / 50.8	7.67 / 5.91 [‡]
Unigram	91.1 / 59.8	24.1 / 24.2	97.0 / 80.0	24.3 / 23.1 [‡]	99.1 / 80.2	7.68 / 5.80 [‡]
EXPGumbel	98.8 / 60.1	23.7 / 23.2 [‡]	96.3 / 56.9	23.7 / 20.9 [‡]	97.8 / 58.2	4.89 / 5.02 [‡]
K-SemStamp	98.6 / 60.7	24.7 / 24.4	98.5 / 65.6	23.4 / 22.5 [‡]	98.5 / 60.7	7.67 / 5.78 [‡]
OURS						
Vanilla	93.7 / 98.6	24.5 / 24.4	83.3 / 98.4	24.0 / 23.1 [‡]	66.6 / 90.3	7.63 / 5.85 [‡]
ELI5	99.9 / 100.0	26.6 / 24.9 [†]	90.1 / 98.4	24.4 / 23.1 [‡]	82.1 / 92.5	7.66 / 5.95 [‡]
MN	98.6 / 99.4	25.4 / 24.2 [‡]	96.7 / 98.4	26.4 / 23.8 [‡]	87.4 / 94.3	7.53 / 5.83 [‡]
IELTS	98.5 / 99.7	25.3 / 24.9 [†]	89.0 / 98.7	24.6 / 23.2 [‡]	85.6 / 96.5	7.65 / 6.10 [‡]

Table 12: Full detection and task performance results before and after the Dipper paraphrasing attack, for Llama-3 and Qwen3 across all three benchmarks. Each cell reports *Before / After* the attack. Task performance columns (ROUGE-L, Judge) serve to verify that paraphrasing preserves semantic content. Boxes highlight out-of-domain evaluations (train task $\neq$ eval task) for PUPPET rows. [†] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

out-of-domain ones—the same structure as before the attack. These observations suggest that PUPPET learns detection-relevant features that are not merely surface-level and thus survive rewriting, while also highlighting that the detection result after paraphrasing depends partly on how sensitive each detector’s features are to surface-level perturbations.

Consistent task quality degradation. In contrast to the detector-dependent detection results, task performance consistently declines after paraphrasing across all detectors and both models. As discussed above, this reflects a property of Dipper—a T5-based model whose output quality is substantially lower than that of Llama-3 or Qwen3—rather than any interaction with the training setup or detector choice.

E Training Efficiency: Qwen3-8B

This section presents sample-efficiency curves for Qwen3, complementing the Llama-3 results shown in Figure 4 in Section 5.3.

Figure 6 shows detection performance (AUROC, top row) and task performance (bottom row) as a function of training samples seen for Qwen3 across all three benchmarks: ELI5 (left), Multi-News (center), and IELTS (right). Dashed and dotted lines indicate KGW and SynthID performance, respectively.

Detector	Method	ELI5		Multi-News		IELTS	
		AUROC	ROUGE-L	AUROC	ROUGE-L	AUROC	Judge
Llama-3							
Fake-SpotAI	Vanilla	100.0 / 100.0	22.9 / 23.3	100.0 / 97.8	26.5 / 23.3	99.7 / 95.9	6.77 / 5.46
	PUPPET	100.0 / 100.0	25.2 / 24.9	100.0 / 99.9	26.1 / 23.2 [‡]	99.9 / 96.1	7.32 / 5.93 [‡]
MAGE	Vanilla	100.0 / 99.9	22.9 / 23.3	97.2 / 86.9	26.5 / 23.3	89.4 / 94.0	6.77 / 5.46
	PUPPET	100.0 / 99.7	24.5 / 24.3 [‡]	95.4 / 84.9	29.2 / 25.2 [‡]	94.2 / 95.1	7.31 / 5.34 [‡]
Qwen3							
Fake-SpotAI	Vanilla	100.0 / 100.0	24.5 / 24.4	100.0 / 98.9	24.0 / 23.1	98.9 / 94.5	7.63 / 5.85
	PUPPET	100.0 / 100.0	25.7 / 25.0 [‡]	100.0 / 99.6	24.1 / 22.7 [‡]	99.6 / 94.4	7.63 / 6.02 [‡]
MAGE	Vanilla	100.0 / 99.9	24.5 / 24.4	97.5 / 89.1	24.0 / 23.5	91.3 / 94.5	7.63 / 5.85
	PUPPET	100.0 / 99.7	25.3 / 24.8 [‡]	97.4 / 88.3	25.9 / 23.5 [‡]	95.3 / 94.6	7.85 / 5.49 [‡]

Table 13: In-domain detection and task performance before and after the Dipper paraphrasing attack, for PUPPET trained with FakeSpotAI and MAGE, across Llama-3 and Qwen3. Each cell reports *Before / After* the attack. Each PUPPET row reports the in-domain result for each benchmark (i.e., the model trained on that task and evaluated on the same task). Task performance columns (ROUGE-L, Judge) serve to verify that paraphrasing preserves semantic content. Before-paraphrase values are taken from Table 11. [‡] and [‡] denote significant (paired t-test ($\alpha = 0.05$)) improvement and degeneration on task performance, respectively.

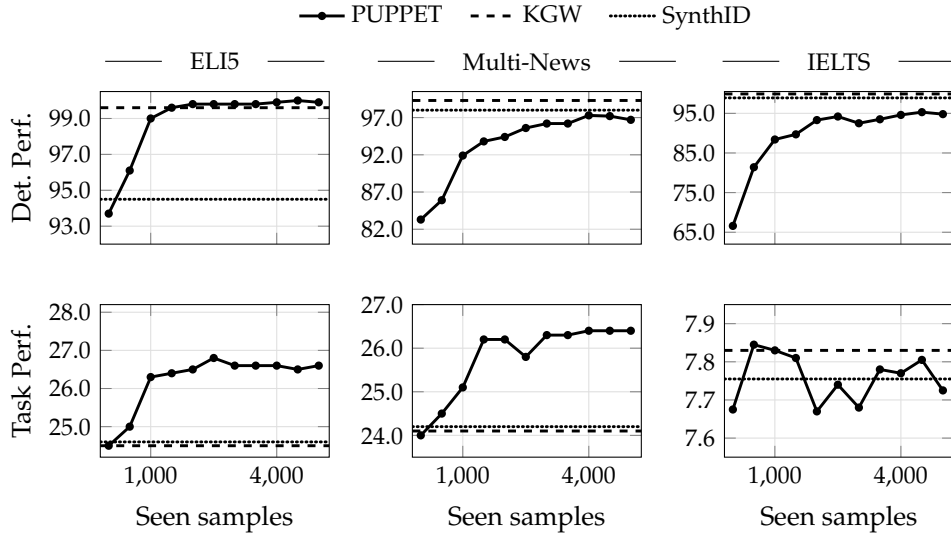


Figure 6: Detection performance (Det. Perf., AUROC; top row) and task performance (Task Perf., ROUGE-L for ELI5 and Multi-News, judge score for IELTS; bottom row) as a function of training samples seen for Qwen3 across all three benchmarks. Dashed and dotted lines indicate KGW and SynthID baselines, respectively.

Qwen3 exhibits a similar overall pattern to Llama-3: both detection and task performance converge within only a few thousand DPO training examples. On IELTS, where the Vanilla detection baseline is considerably lower (66.6, compared to 83.3 on Multi-News), detection performance requires more samples before approaching its ceiling. The task performance curve for Qwen3 on IELTS is notably unstable and fails to converge to a clear plateau within the observed range, which we attribute to the difficulty of detection by OpenAI Detector in this setting (Vanilla Qwen3-8B achieves only 66.6 AUROC on IELTS).

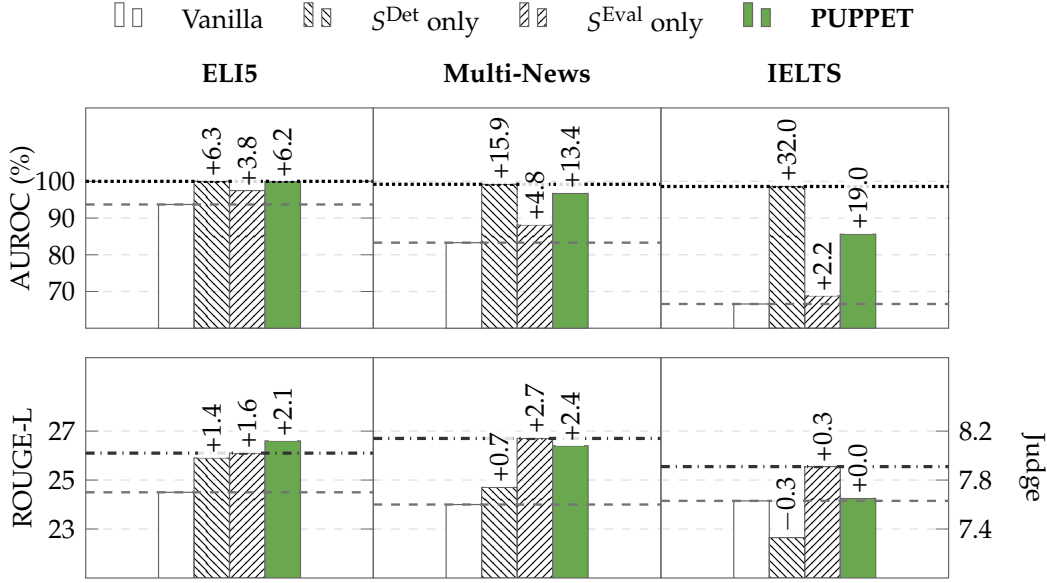


Figure 7: Reward ablation (Qwen3). Bar labels: point-wise changes from Vanilla. In the AUROC panels, dotted lines mark the S^{Det} -only ceiling; in the task panels, dash-dotted lines mark the S^{Eval} -only ceiling; dashed lines mark the Vanilla baseline in all panels. ROUGE-L (left) and Judge (right) task axes are proportionally scaled.

F Reward Composition Ablation: Qwen3-8B

Figure 7 reports reward composition ablation results for Qwen3, complementing the Llama-3 results in Figure 5.

The overall pattern is consistent with Llama-3: S^{Det} only achieves large AUROC gains at the cost of task performance, S^{Eval} only improves task without lifting detection, and PUPPET improves on both axes. On ELI5 and Multi-News, PUPPET recovers—and in some cases exceeds—the single-objective gains: 84.3–98.4% of the S^{Det} -only detection gain and 88.9–131.3% of the S^{Eval} -only task gain.

A notable difference from Llama-3 appears on IELTS, where PUPPET’s detection gain recovery drops to 59.4% (vs. 95.3% for Llama-3). This reflects a harder trade-off: the base model’s IELTS AUROC is substantially lower (66.6 vs. 87.0 for Llama-3), and the task-performance objective is weaker relative to the detection signal, making it harder for concurrent training to fully capture the detection improvement with a fixed $\alpha = 0.5$.

G Full Model-Attribution Results

Table 14 reports model-attribution AUROC from the Qwen3 perspective, complementing the Llama-3 results in Table 5.

At the vanilla baseline, the detector achieves AUROC of 41.01 on ELI5 (below chance from this perspective) and 27.74–22.29 on Multi-News and IELTS—the mirror image of the Llama-3 baseline, confirming that the asymmetry is an artifact of the uncontrolled detector rather than a meaningful signal in either direction.

Applying PUPPET to Qwen3 substantially raises AUROC against Vanilla Llama-3 on ELI5 (87.98) and IELTS (71.09), though the gain on Multi-News (59.72) is more modest. These results are consistently lower than the PUPPET Llama-3 versus Vanilla Qwen3 counterparts (94.47, 96.94, 99.14; Table 5). We attribute this asymmetry to the baseline machine-class likelihood distributions: Vanilla Llama-3 already occupies a high machine-class likelihood

Target	Contrast	ELI5	Multi-News	IELTS
Vanilla Qwen3	Vanilla Llama-3	41.01 (−0.23)	27.74 (−0.82)	22.29 (−0.96)
PUPPET Qwen3	Vanilla Llama-3	87.98 (1.22)	59.72 (0.31)	71.09 (0.78)
PUPPET Qwen3	Vanilla Qwen3	90.97 (1.51)	81.18 (1.12)	91.17 (1.97)
PUPPET Qwen3	PUPPET Llama-3	45.46 (−0.22)	18.37 (−1.21)	12.79 (−1.13)

Table 14: Qwen3 model-attribution AUROC (%) (OpenAI Detector); treating *Target* as the positive class and *Contrast* as the negative. Parentheses: Cohen’s d ; models are trained per task. Negative d indicates the Target’s machine-class likelihood lies below the Contrast’s.

region, whereas Vanilla Qwen3 lies in a lower one (Table 1 for Llama-3 and Table 8 for Qwen3). When PUPPET shifts Qwen3’s distribution upward, it reaches a level comparable to Vanilla Llama-3 rather than surpassing it, leaving the two distributions insufficiently separated—most strikingly on Multi-News, where $d = 0.31$. By contrast, the analogous gap for PUPPET Llama-3 versus Vanilla Qwen3 is $d = 3.18$ on the same benchmark, explaining the markedly higher AUROC in Table 5.

That the upward shift does occur for Qwen3 is independently confirmed by the self-comparison: PUPPET Qwen3 is reliably distinguishable from Vanilla Qwen3 across all benchmarks ($d = 1.12$ – 1.97), at a magnitude consistent with the analogous Llama-3 shift (cf. Table 5).

In the condition that both models are trained with PUPPET, AUROC from the Qwen3 perspective (45.46, 18.37, 12.79) is the arithmetic complement of the values from the Llama-3 perspective (54.54, 81.64, 87.22; Table 5): swapping the positive and negative classes reflects AUROC around 50, so these two rows convey the same information presented from opposite sides.